

Building A Highly Available Nginx Reverse-Proxy Using

Heartbeat

Last month we discussed how to set up a highly available cluster of Web servers that are load balanced using nginx. One shortcoming in that set-up was the reverse-proxy server itself, which on crashing, will cause the entire Web server cluster to go down. Therefore, we would need to build high-availability in the reverse-proxy server itself.

A cluster in computing is a term used to describe a group of closely linked computers often appearing as a single entity to the outside world. There are various types of clusters -- high-availability (HA) clusters, load-balancing clusters, compute clusters or high performance computing (HPC) clusters and grids.

An HA cluster is also known as a failover cluster and is typically meant to improve service availability rather than performance, by using redundant nodes. There are many models of HA cluster configuration such as active-passive, active-active, N+1, N+M, N-to-1 and N-to-N.

Load-balancing clusters distribute the workload evenly among various redundant nodes. There are various different algorithms, using which the load-balancer distributes the load among member servers.

HPC clusters are used for highly CPU-intensive compute jobs. There are various types of compute clusters. The common distinguishing factor is the coupling of the compute nodes. Typically, there are specialised scientific applications that run on these clusters. The applications are built

using libraries supporting parallel processing. A popular example of a compute cluster is the Beowulf Clusters. These are built using commodity hardware and run on FOSS systems like FreeBSD or GNU/Linux. Typically, a Beowulf cluster uses either MPI (Message Passing Interface) or PVM (Parallel Virtual Machine) libraries that allow a programmer to divide a task among the nodes of the cluster, and then recollect and assemble the results later on.

A grid is a special class of compute clusters with possibly heterogeneous nodes that are not so tightly coupled with each other. All nodes in the grid target single problems that require a great number of CPU cycles and a large amount of data. A grid typically divides the entire computation work into jobs that are independent and do not share data with each other. The intermediate results of one job in the grid do not affect the other jobs running on other nodes.

Heartbeat is a piece of software from "The High Availability Linux" project, which provides high-availability clustering solutions for a wide range of *nix operating systems, including (though not limited to) GNU/Linux, FreeBSD and OpenBSD.